Method of Discrete Singularities in the Accurate Modeling of a Reflector Beam Waveguide

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Abstract - **We study the *E* and *H*-polarized beam wave guidance and scattering by finite-size reflectors modeling quasioptical beam waveguides. The incident complex-source-point (CSP) field simulates a beam generated by a small-horn aperture. Numerical solution is obtained from the coupled singular integral equations (SIEs) discretized by using new quadrature formulas of interpolation type.**

Index Terms - **Beam waveguides, focusing, boundary integral equations, scattering, diffraction, numerical analysis**

I. Introduction

Since the 1930's, metallic parabolic reflectors have been used as high-gain quasioptical antennas [1]. Parabolic shape provides a well-known property of collimating the incident beam of a small horn placed in the geometrical focus. As a further development of quasioptical techniques, reflector beam waveguides were proposed in the 1960's for guiding the millimeter (mm) waves along a chain of reflectors [2,3]. Here, the basic shape is elliptic one known to provide a necessary phase correction. Today reflector beam waveguides are used, e.g., in the heating of plasma in controlled nuclear fusion machines with mm waves generated by high-power gyrotrons [4,5]. Electromagnetic modeling of reflectors is usually done with "quasioptical" methods, the main being Geometrical and Physical Optics supplemented with Geometrical and Uniform Theories of Diffraction [6-8]. However, these methods are based on the ray tracing and fail to characterize interaction effects and resonances. For a more accurate modeling, full-wave methods are mandatory. However, widely popular today Finite-Element and Finite-Difference field solvers require prohibitively large computer resources when applied even to a single reflector larger than 10 wavelengths in size, in open domain.

All this shows that economic and reliable tools for the modeling of reflector-type scatterers are still in demand. A general way to build such tools is to use the IE approach. Here, the crucial point is development of an efficient discrete model, i.e. a fast and convergent numerical algorithm having controlled accuracy. Since the 1990's, discrete models based on the analytical preconditioning, i.e. conversion of a singular IE for the surface current function to a Fredholm second kind matrix equation are known [9]. We present here the basic ideas of another discrete model based on the Method of Discrete Singularities (MDS), and review the results of accurate modeling of quasioptical reflector antennas and waveguides.

II. Problem formulation

The geometry of a generic 2-D two-reflector structure viewed as an elementary cell of a beam waveguide can be seen in Fig.1.

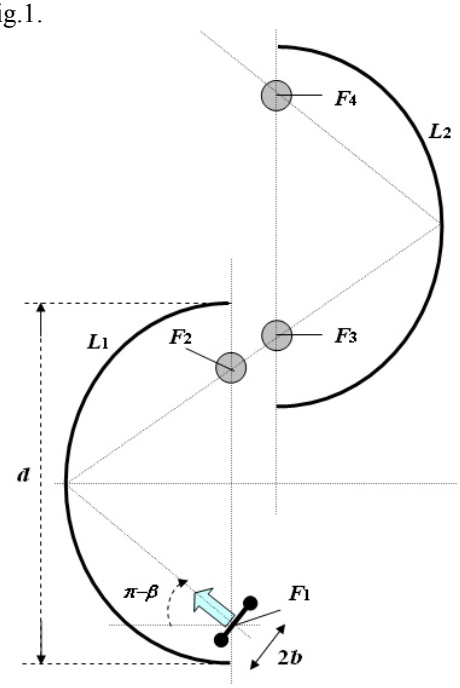

Fig. 1 Geometry of a two-reflector scattering structure

Reflectors, which are most frequently elliptic, are assumed to be perfectly electric conducting (PEC) and have zero thickness. The feed is a line current placed at the complex-valued source point and has time dependence $e^{-i\omega t}$ omitted in the analysis (see also [10]). The field generated by such a CSP feed can be characterized by the *z*-component of the electric field, which is given by

$$U_0(\vec{r}) = H_0^{(1)}\left(k\left|\vec{r}-\vec{r}_c\right|\right), \quad \vec{r}_c = \vec{r}_0 + i\vec{b}\ , \ \vec{r}_0 = (x_0, y_0), \quad \vec{b} = (b\cos\beta, b\sin\beta) \qquad \textbf{(1)}$$

where $k = \omega / c = 2\pi / \lambda$. Function (1) is a rigorous solution to the Helmholtz equation. It can be shown that (1) simulates a beam wave and reaches maximum in the direction $\phi = \beta$ for all *r*. Besides, it has two branch points at $(x_0 \pm b\cos\beta,\ y_0 \mp b\sin\beta)$ - thus to single out a unique value of $U_0(x, y)$ one has to join them with a branch cut *B* of the length 2*b* in the real space that does not intersect the contours of reflectors, $L_{1,2}$. This cut can be considered as a

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model of the real horn aperture, and the greater *kb*, the narrower the CSP beam.

The field (1) is taken as the incident one, thus the total filed is a sum, $U = U_{sc} + U_0$, where $U_{sc}(x, y)$ is the secondary field scattered by reflectors. To simulate an antenna in 2-D, we study the scattering of *E* and *H*-polarized waves (1) by two curved PEC strips as electromagnetic boundary-value problem. It requests U_{sc} to solve the Helmholtz equation off $L_{1,2}$ and satisfy (a) the PEC boundary condition on $L_{1,2}$, (b) the edge conditions at strip endpoints and (c) the radiation condition. This problem is reduced to a set of two coupled SIEs of the first kind for the surface currents induced on the strips, $j_{1,2}(s)$:

E-polarization:

$$\sum_{q=1,2} \frac{i}{4} \int_{L_q} H_0^{(1)}\left(k\left|\vec{r}_q(s) - \vec{r}_p(s_0)\right|\right) j_q(s) ds = -U_0\left(\vec{r}_p(s_0)\right)\Big|_{L_p}$$
$$p = 1,2 \qquad \textbf{(2)}$$

H-polarization:

$$\sum_{q=1,2} \frac{1}{2\pi} \frac{\partial}{\partial n_p} \int_{L_q} j_q(x) \frac{\partial}{\partial n_q} H_0^{(1)}\left(k\left|\vec{r}_q(s) - \vec{r}_p(s_0)\right|\right) ds =$$
$$= -\frac{\partial}{\partial n_p} U_0\left(\vec{r}_p(s_0)\right)\Big|_{L_p}, \quad p = 1,2 \qquad \textbf{(3)}$$

Note that (2) has logarithmic singularities in the kernels while (3) is a set of hyper-singular equations.

III. Method of Discrete Singularities

Applying contour parameterization *x(t)*, *y(t)* one can transform SIE (2) to another, Cauchy-singular IE set with supplementary conditions. For brevity, here we present such equation for a single reflector problem:

$$\frac{1}{\pi}\int_{-1}^{1} \frac{v(t)}{t - t_0} \frac{dt}{\sqrt{1-t^2}} + \frac{1}{\pi}\int_{-1}^{1} K(t, t_0) v(t) \frac{dt}{\sqrt{1-t^2}} = f(t_0),$$
$$\frac{1}{\pi}\int_{-1}^{1} M(t) v(t) \frac{dt}{\sqrt{1-t^2}} = c \qquad \textbf{(4)}$$

For a single reflector, SIE (3) is converted to:

$$\frac{1}{\pi}\int_{-1}^{1} \frac{v(t)}{(t_0 - t)^2} \sqrt{1-t^2}\, dt - \left(\frac{k|L|}{8}\right)^2 \frac{1}{\pi}\int_{-1}^{1} v(t) \ln|t_0 - t| \sqrt{1-t^2}\, dt +$$
$$+ \frac{1}{\pi}\int_{-1}^{1} v(t) N(t_0, t) \sqrt{1-t^2}\, dt = g(t_0) \qquad \textbf{(5)}$$

Here *K, M, N, f, g* are smooth functions, *c* is a known constant, and *v(t)* is new smooth unknown function. IEs (4,5) are further discretized by using the quadrature formulas of interpolation type with the nodes in the nulls of the Chebyshev polynomials of the first kind: $t_i^n = \cos\left[\pi(2i-1)/2n\right]$, $T_n(t_i^n) = 0$, $i = 1,2,...n$, and the second kind $t_{0j}^n = \cos(\pi j/n)$, $U_{n-1}(t_{oj}^n) = 0$, where $j = 1,2,...n-1$. As a result, for (4) we get a set of the following linear equations:

$$\frac{1}{n}\sum_{i=1}^{n} \frac{v(t_i^n)}{t_i^n - t_{0j}^n} + \frac{1}{n}\sum_{i=1}^{n} K(t_i^n, t_{0j}^n) v(t_i^n) = f(t_{0j}^n), \quad (j = 1,2,...n-1)$$
$$\frac{1}{n}\sum_{i=1}^{n} M(t_i^n) v(t_i^n) = c, \quad (j = n), \qquad \textbf{(6)}$$

and for (5)

$$\frac{1}{n}\sum_{\substack{q=1\\ q\neq j}}^{n-1} v_{n-2}(t_{0q})\left(1 - t_{0q}^2\right)\left\{\frac{1-(-1)^{j+q}}{(t_{0j} - t_{0q})^2} + \lambda^2 S_n(t_{0j}, t_{0q}) + N(t_{0j}, t_{0q})\right\} +$$
$$+ v_{n-2}(t_{0j})\left\{-\frac{n}{2} + \frac{(1-t_{0j}^2)}{n}\left(\lambda^2 S_n(t_{0j}, t_{0j}) + N(t_{0j}, t_{0j})\right)\right\} = g(t_{0j}) \qquad \textbf{(7)}$$

Here $(k|L|/8)^2 = \lambda^2$,

$$S_n(t_{0j}, t_{0q}) = \ln 2 + 2\sum_{l=1}^{n-1} \frac{1}{l} T_l(t_{0j}) T_l(t_{0q}) + \frac{(-1)^{j+q}}{2n},$$

and $j = 1,2,...n-1$.

Generalization of (4) – (7) to multi-reflector case is straightforward. This method of numerical solution has been developed recently; its essentials can be found in [11,12]. It enables one to study the effects of beam wave guidance and scattering for reflector antennas and waveguides with high accuracy. Near and far field patterns, surface currents, and also focusability and gain can be readily computed for various reflectors, feed locations, etc.

IV. Numerical results

Near field focusing with elliptic reflectors

Consider a single elliptic reflector, whose contour is $L = \left\{(x, y) \in R^2 : (x/a_1)^2 + (y/a_2)^2 = 1\right\}$, and a CSP feed placed is in the geometrical focus F_1 (Fig. 2). This geometry is interesting for the near-field focusing applications such as mm-wave plasma heating and microwave laser pumping.

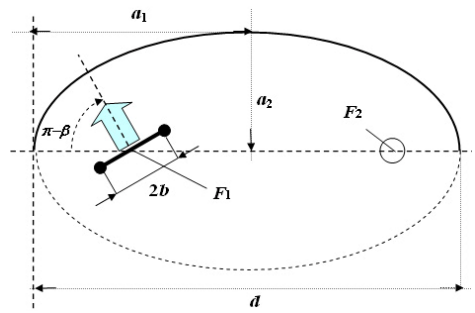

Fig. 2 Geometry of single elliptic reflector fed by a CSP feed.

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Fig.3 illustrates how the reflected rays gather at the other focus F_2.

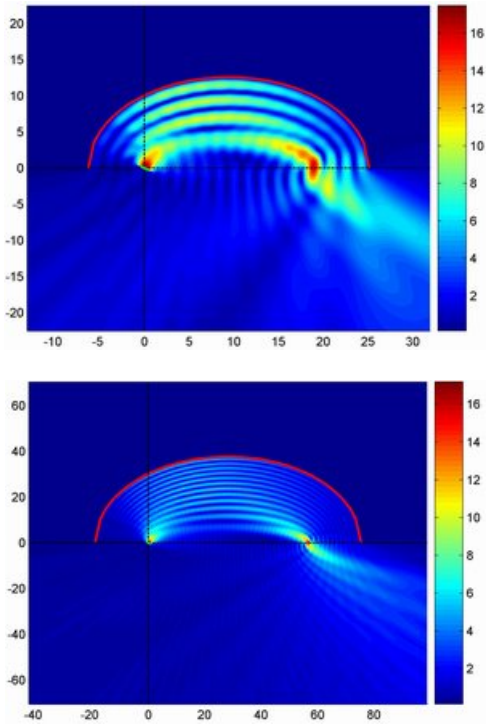
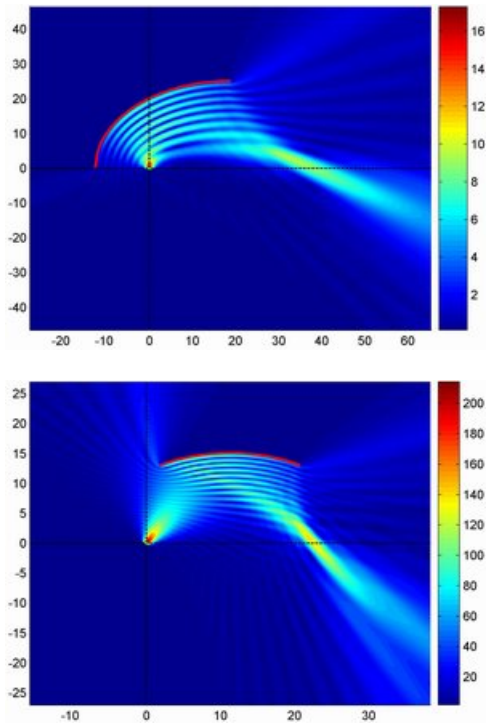
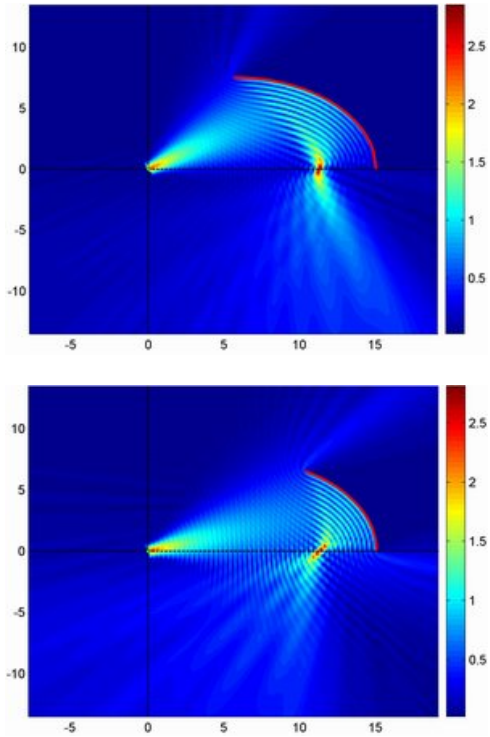


Fig. 3 Near fields of half-ellipse focusers having d=10λ, kb=2 (top) and d=30λ, kb=2 (bottom).

Elliptic reflector focusers are also known in practical applications; we show here several examples for the various d/λ and kb values. In all cases, the high-intensity field spot in the other focus is seen clearly (Figs. 4). If a numerical optimization is a goal of simulation, the focusability can be accurately quantified and used as an aim function.





F*ig*. 4 Near fields of single-reflector offset focusers having: quarter-ellipse shape with d=12λ, kb=2 (top); quarter-ellipse with d=15λ, kb=5 (middle-top); quarter-ellipse with d=19λ, kb=10 (middle-bottom); and 1/6-ellipse with d=12λ, kb=10 (bottom).

Elliptic reflector beam waveguide

We have also made preliminary simulations for a 2-D reflector beam waveguide as shown in Fig.1. Suppose that there are two identical elliptic-arc PEC reflectors facing each other however shifted along the both axes. The primary field is again a beam-like directive field of a CSP line-current feed whose aperture is centered at a focus of the nearest reflector. Such a structure can be considered as a model of elementary cell of realistic mm-wave reflector beam waveguide used to transmit high-power waves over a distance in free-space environment [2-5]. In this case, the modeling deals with two coupled SIEs (2) and results in two coupled matrix equations of the same type and properties as the one obtained for a single-reflector structure. Figs. 5 and 6 show the evolution of the near field portraits if the geometry varies from two non-confocal half-elliptic reflectors to confocal ones of smaller size. They also demonstrate that one can cut off the "unnecessary" parts of the reflectors so far as the edge illumination is kept lower than -10 dB with respect to the center. The pattern in Fig. 6 demonstrates a perfect guidance of the wave beam, whose focal spot repeats itself at the most distant focus of the second reflector almost without distortions.

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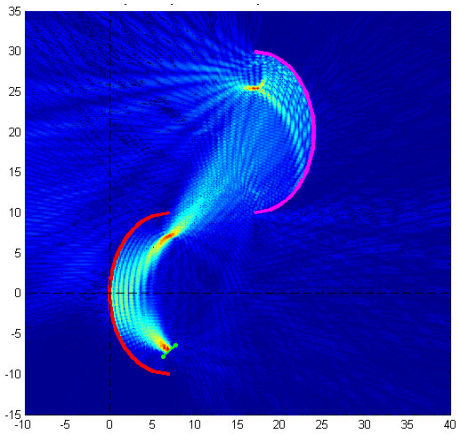

*Fig.*5 Near field for two **non-confocal** half-ellipses fed by a CSP feed, d=12λ, kb=9, β=140°.

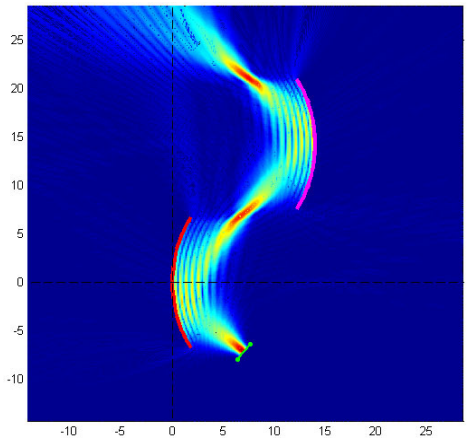

*Fig.*6 Near field of two **confocal** elliptic reflectors of optimal size, d=20λ, kb=9, β=140°.

Therefore no surprise that if we use a third reflector placed confocally with the second one, we provide a further guidance (Fig. 7).

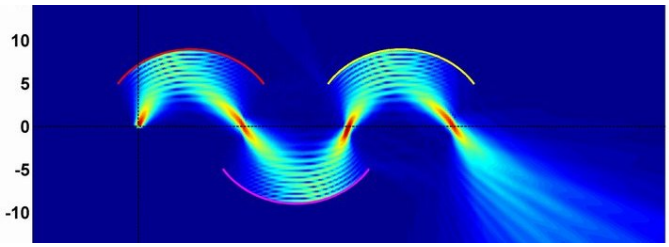

*Fig.*7 Near field of three **confocal** elliptic reflectors of optimal size, d=20λ, kb=7, β=64°.

VII. Conclusion

With the aid of the presented numerical method based on MDS one can compute 2-D models of real-life reflector antennas of arbitrary shape mostly in minutes, with desired accuracy and for both polarization cases. For moderate computing resources, even household, like for example 2GHz, AMD AthlonXP desktop computer with 512Mb RAM, the time for computing the near E-field of a d=100λ reflector for 250x250 field points is approximately 2-3 minutes, and can be greatly reduced on more powerful modern multi-processor computer facilities. Thus, the electromagnetic performance of beam waveguides as well as multi-reflector antennas can be controlled by drawing and analyzing the field patterns, directivity plots, etc., for a multitude of parameters like kb, d/λ, etc. Besides, MDS has a potential to be extended to the 3-D reflectors, at least rotationally symmetric ones. Finally, it is relatively straightforward to modify it for imperfect reflectors, either impedance, dielectric, or resistive. Due to a lucky combination of high efficiency and controlled accuracy, the method can be used as a core of the computer optimization software based on the gradient-type approach or on the genetic-algorithm one.

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